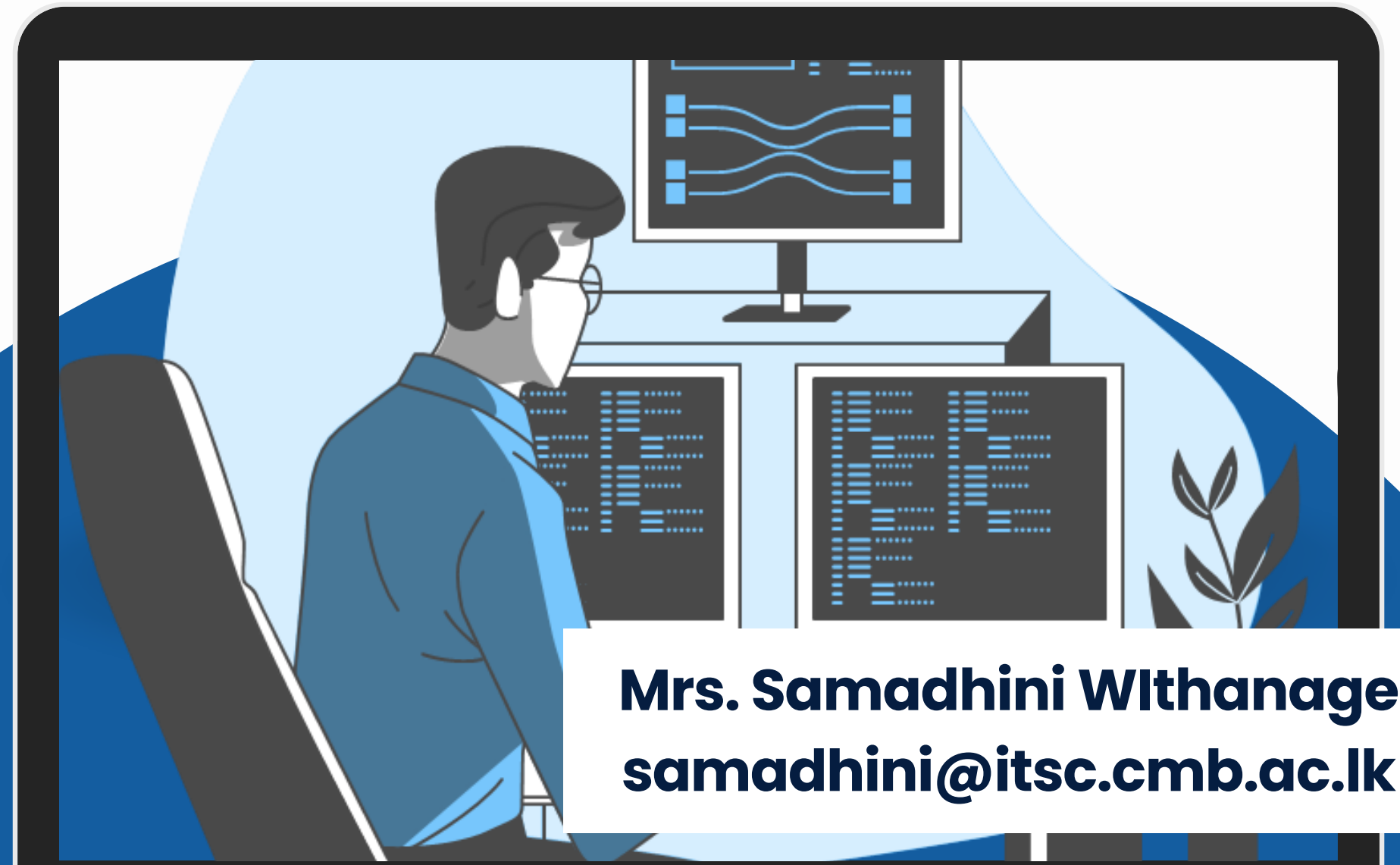


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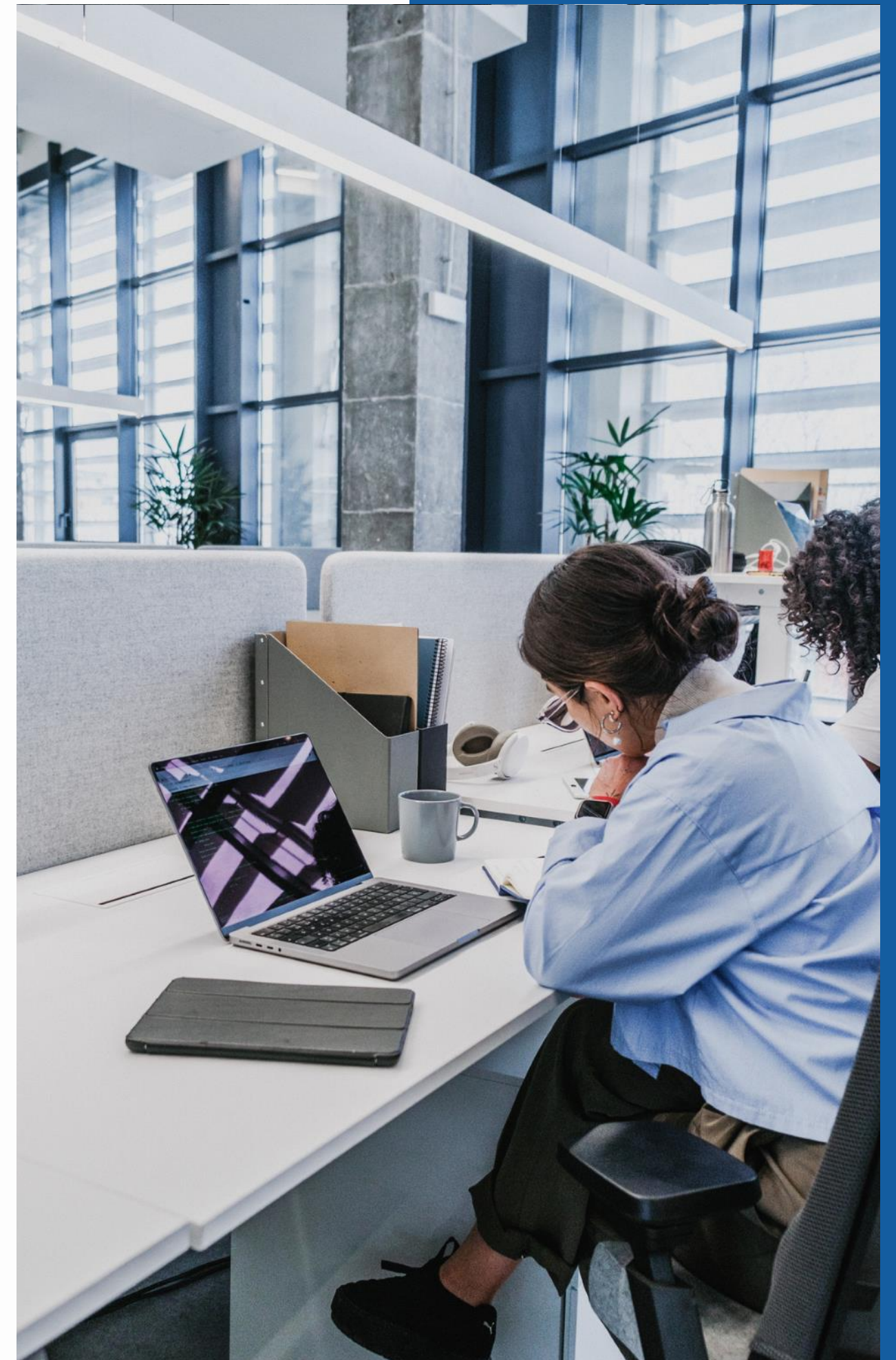
# Advanced Programming Techniques



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# Course Outline

- ▶ Introduction to Programming
- ▶ Java Basic
- ▶ OOP Concepts
- ▶ UML & Object-Oriented Design
- ▶ File Handling & Exception Handling
- ▶ Garbage Collection
- ▶ Version Control (Git/Bitbucket)
- ▶ Coding Standards & Best Practices
- ▶ Software Design Patterns
- ▶ AI-Assisted Programming



# Learning Outcomes

At the end of this course module the students will be able to

01

Maintain a good proficiency in basic programming concepts

02

Understand the main programming paradigms of procedural and object-oriented programming and their usages

03

Show a good understanding of Advanced programming concepts such as File handling, Database programming and GUI application development

04

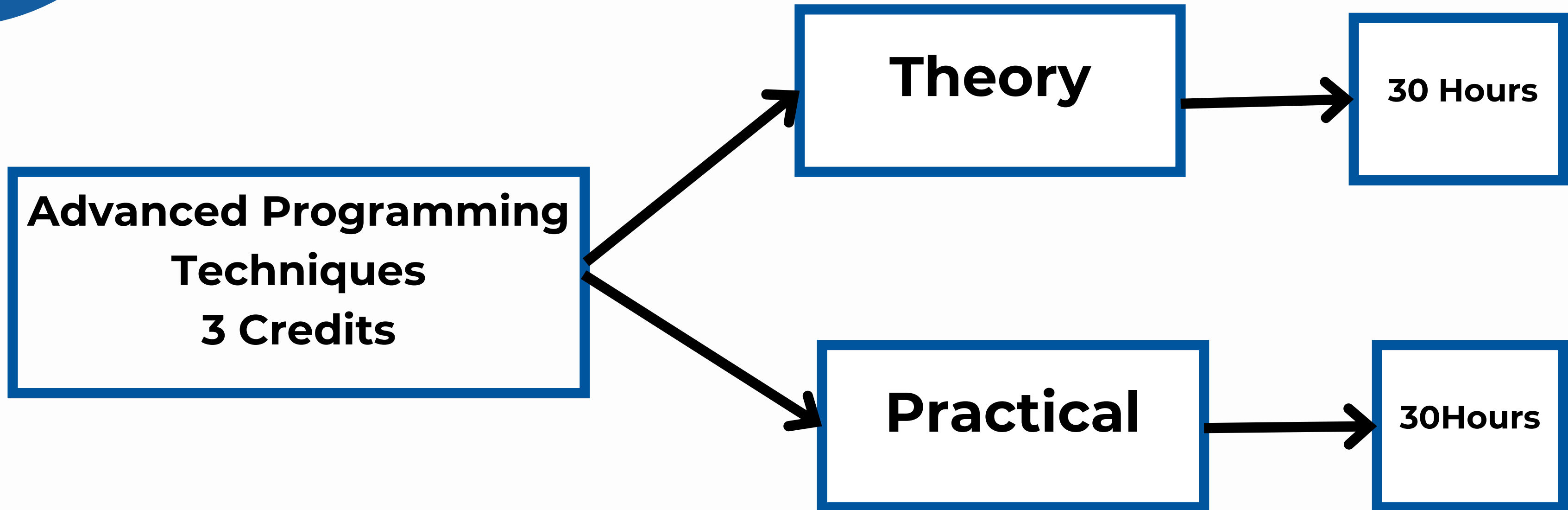
Possess a good knowledge about different Software design patterns used in the industry with coding standards and best practices

05

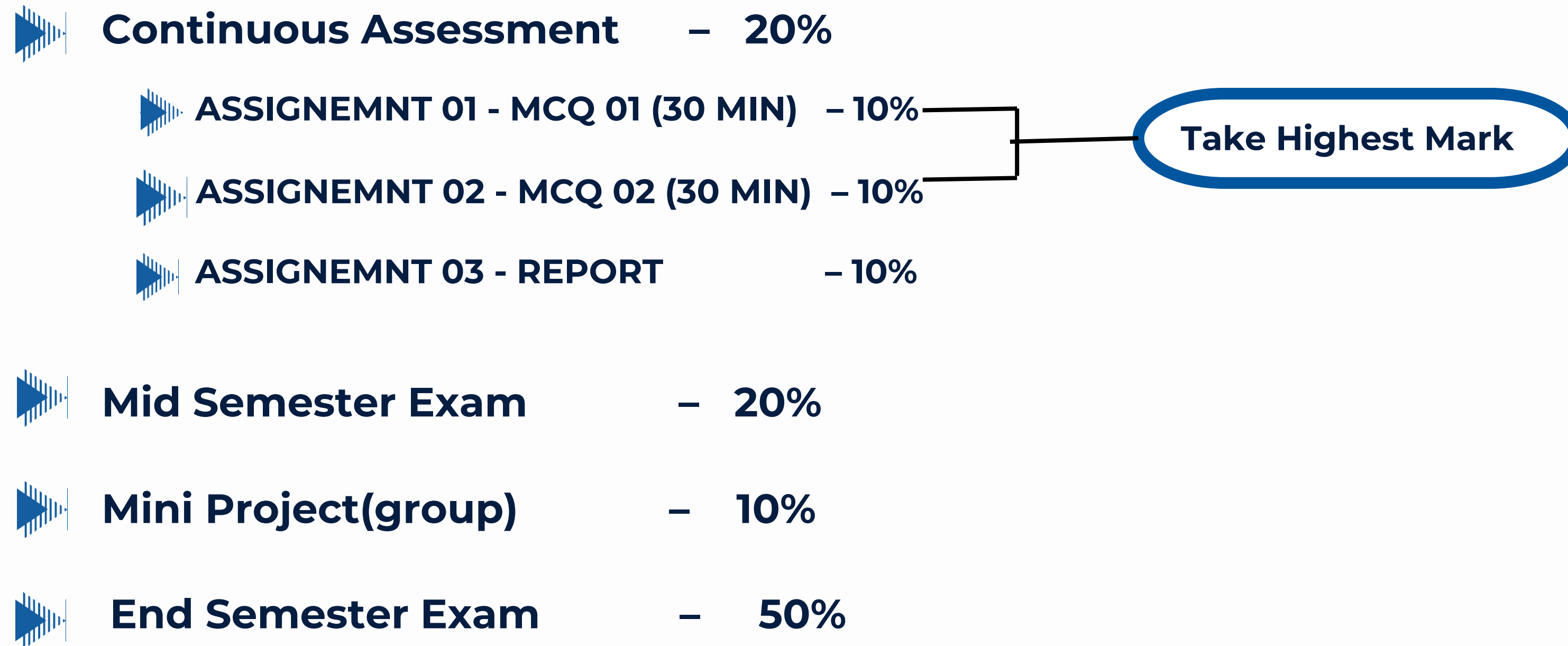
Demonstrate the ability to design(model) and implement a practical solution utilizing the programming concepts learned through the module



# Course Structure



# Evaluation Criteria



# Recommended Reading



Java: The Complete Reference, Ninth Edition - Herbert Schildt



Design Patterns: Elements of Reusable Object-Oriented Software - Book by Erich Gamma, John Vlissides, Ralph Johnson, and Richard Helm



Clean Code: A Handbook of Agile Software Craftsmanship 1st Edition -by Robert C. Martin

The background is a blurred office scene with people working at desks, overlaid with a semi-transparent blue filter. A white rectangular box is centered on the image, containing the text "ANY QUESTIONS ?" in a bold, white, sans-serif font. The word "ANY" is on the top line, and "QUESTIONS ?" is on the bottom line.

**ANY  
QUESTIONS ?**



**THANK YOU!**

